

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every ~7 Months

50% task-completion time horizon doubles every ~207 days ($R^2 = 0.97$)

Thomas Kwa, Ben West, Joel Becker, et al. — METR

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AI Safety Researchers · Frontier Lab Leadership · Policy Analysts

A New Metric: 50% Task-Completion Time Horizon

Human-grounded measure of real-world AI autonomous capability

1. Assemble Task Suite

170 diverse software / research tasks spanning 1 second to 8 hours of skilled human effort

2. Measure Humans & AI

Time skilled human professionals; measure AI agent success rates on identical tasks

3. Fit Logistic Model

Estimate task duration at which each AI achieves 50% success — the time horizon

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Three benchmark sources • 800+ human baselines • 2,529 hours collected

HCAST • 97 tasks

1 min – 30 hrs
Diverse software engineering tasks

RE-Bench • 7 tasks

~8 hrs each
Difficult ML research engineering

SWAA • 66 tasks

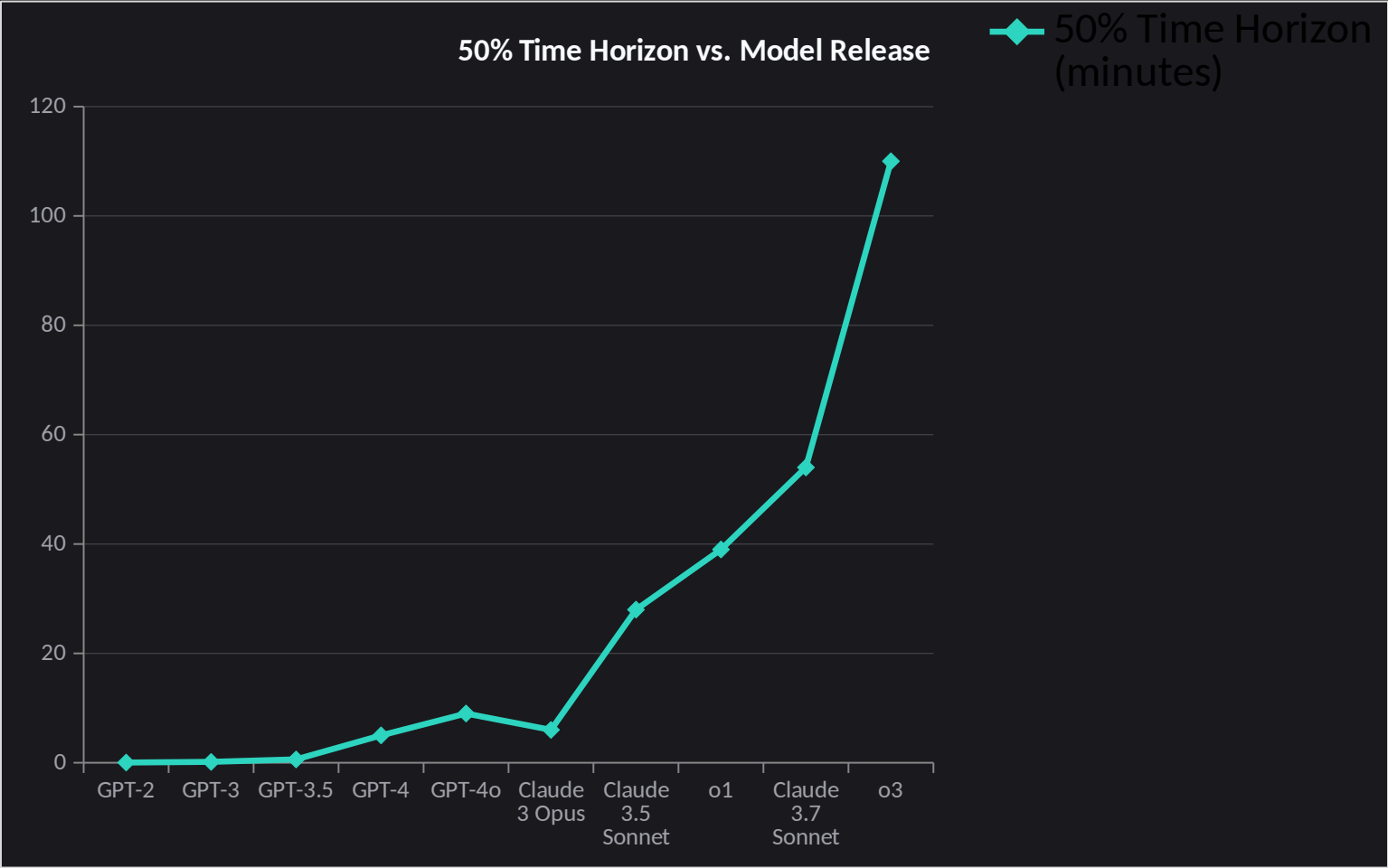
1 sec – 30 sec
Short single-step atomic actions

Example Tasks by Human Duration

Task Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: 'Which file is a shell script?'
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30× speedup of a backtesting tool

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years

Exponential growth with ~207-day doubling time • $R^2 = 0.97$



Key Statistics

Doubling time: ~207 days (~7 months)

$R^2 = 0.97$ (6 years of model releases)

2023–2025 growth: ~20% faster than 2019–2025

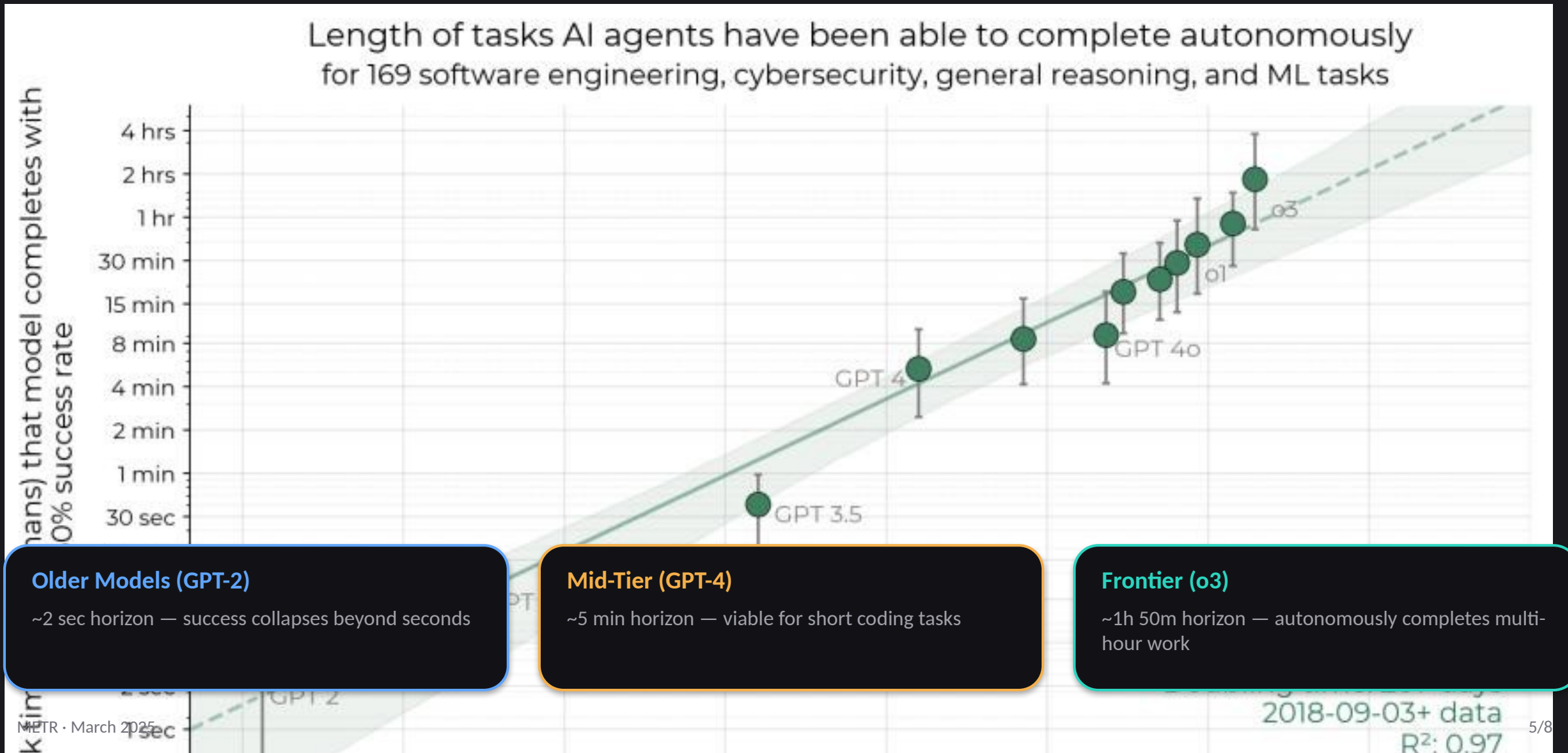
Model Milestones

GPT-2 (2019)	~2 seconds
GPT-4 (2023)	~5 minutes
Claude 3.5 Sonnet (2024)	~28 minutes
o1 (2024)	~39 minutes
Claude 3.7 Sonnet (2025)	~54 minutes
o3 (2025)	~1 hour 50 minutes



Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models push the curve rightward — higher success at longer durations



Reasoning, Tool Use, and Error Recovery Drive Time

Capability improvements that extend autonomous execution length

Horizon Growth



Improved Logical Reasoning

Models decompose complex problems into manageable sub-tasks and maintain coherent reasoning chains over longer contexts.



Better Tool Use

Effective utilization of IDEs, debuggers, terminals, and APIs to augment capability beyond pure text generation.



Greater Reliability & Self-Awareness

Models recognize when they are stuck or uncertain, enabling targeted clarification or retry loops.



Ability to Adapt to Mistakes

Error detection and recovery strategies allow agents to backtrack and correct course mid-task.

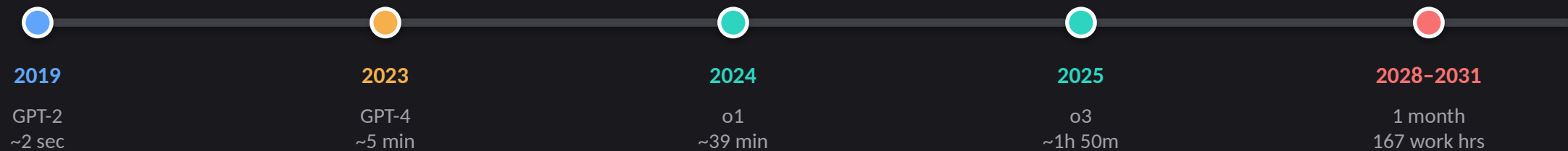
Limitation: Performance is notably lower on less structured, 'messier' tasks. External validity to real-world job distributions remains an open question.

Extrapolation Predicts Month-Long Task Automation by 2028-2031

Naive trend continuation implies automation of many month-long software projects

Forecast: 1-Month Time Horizon (167 Work Hours)

- If the exponential trend continues, AI reaches a >1-month time horizon between mid-2028 and mid-2031.
- This implies automation of many month-long software and research engineering projects.
- The 2023–2025 horizon growth rate is ~20% faster than the overall 2019–2025 rate, suggesting possible acceleration.



Caveats & Uncertainties

- External validity: tasks may not perfectly represent real-world intellectual labor distributions.
- Trend extrapolation assumes no fundamental slowdown in capability scaling or data limitations.
- Supplementary experiments found little evidence that trends are slower on more realistic tasks, but sample sizes were limited.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

Key conclusions for researchers, lab leadership, and policymakers

1

Robust Metric

50% time horizon is a robust, intuitive metric for tracking AI autonomy with $R^2 = 0.97$ over 6 years of model releases.

2

Exponential Pace

Doubling every ~7 months with possible acceleration since 2024 — faster than many governance frameworks assume.

3

Current Frontier

o3 autonomously completes tasks requiring ~2 hours of skilled human effort, up from 2 seconds for GPT-2 in 2019.

4

External Validity

Tasks may not perfectly represent real-world intellectual labor; generalization to messy, unstructured work remains uncertain.

5

Governance Imperative

Month-long automation by 2028–2031 warrants proactive safety standards, eval protocols, and policy preparation today.